

INTRODUCTION

The transition from adolescence to adulthood, a period known as emerging adulthood (ages 18 to 25), represents a critical developmental stage characterized by significant life changes, identity exploration, and a heightened risk for initiating illicit drug use. While traditional research has often examined substances in isolation, modern public health challenges require an understanding of the social and demographic architecture of substance use variety specifically, the total number of different substances an individual has accessed as a proxy for risk-taking behavior and social exposure. The primary aim of this study is to identify how intersecting identities, such as gender and race, alongside institutional engagement like school enrollment, collectively influence the breadth of substance use. Moving beyond simple prevalence rates, this research seeks to determine if certain environments, particularly higher education, serve as a protective "buffer" that mitigates demographic predispositions toward higher drug variety. Understanding these patterns is essential (a) for identifying Protective Factors (Determining if school enrollment acts as a significant deterrent helps clarify how institutional support reduces the likelihood of experimenting with multiple substances.), (b) for addressing Gender-Risk Gaps (previous data suggests males often engage with a higher variety of substances than females; understanding this gap allows for tailored interventions). This investigation is grounded in foundational patterns established by previous national and community-based studies. Prior research, such as that by SAMHSA (2007), has consistently shown that emerging adults utilize illegal drugs at higher rates than any other age group. Furthermore, scholars like Wu et al. (2006) identified significant differences in drug use based on race and school status, noting that students were generally less likely to report use than their non-student peers. Despite these insights, several critical gaps remain in the existing literature: (a) lack of community samples (most research on "club drugs" has focused on high-risk populations, leaving a scarcity of data on general community-based samples of heterosexually identified emerging adults) (b) conflicting gender data (previous studies have offered contradictory evidence regarding gender and poly-drug use; while some suggested females were more likely to use multiple drugs, others found the opposite to be true). By utilizing a multivariable approach and focusing on the total variety of drugs used, this study aims to fill these gaps, providing a comprehensive view of substance-related risk to better inform primary prevention efforts.

METHODS

Here, we will focus on the following 4 variables.

TOTALDRUGS: This is a discrete (count) variable representing the total number of different drugs an individual has used in their lifetime. In this dataset, the values range from a minimum of 0 to a maximum of 12.

GENDER: This is a categorical (binary) variable used to compare differences in drug use patterns between male or female.

RACE: This is a categorical (nominal) variable that identifies the participant's racial or ethnic background. The categories included in the analysis are White/Caucasian, Asian/Pacific Islander (API), Black/African American, and Mixed Race.

SCHOOL (Enrollment Status): This is a categorical (binary/nominal) variable indicating whether a participant is currently enrolled in a college or university. It is coded as "Yes" (currently enrolled) or "No" (not currently enrolled).

We will be using histograms and bar chart to look at the distributions of the different variables and Box plots to look at the relationships between Total drugs and the 3 other variables.

RESULTS

Distribution of Lifetime Drug Use

The average number of drugs used in the lifetime of the 261 young adults surveyed was 2.72 (SD = 2.21), with a range of 0 to 12. The figure1 shows that distribution of Lifetime Drug Use is positively skewed (right skewed) with most participants having used 0 to 4 drugs.

Table 1 Summary statistics of Lifetime Drug Use

MeanTotal drugs	Median	St. Deviation	25% quartiles	75% quartiles
2.72	2	2.21	1	4

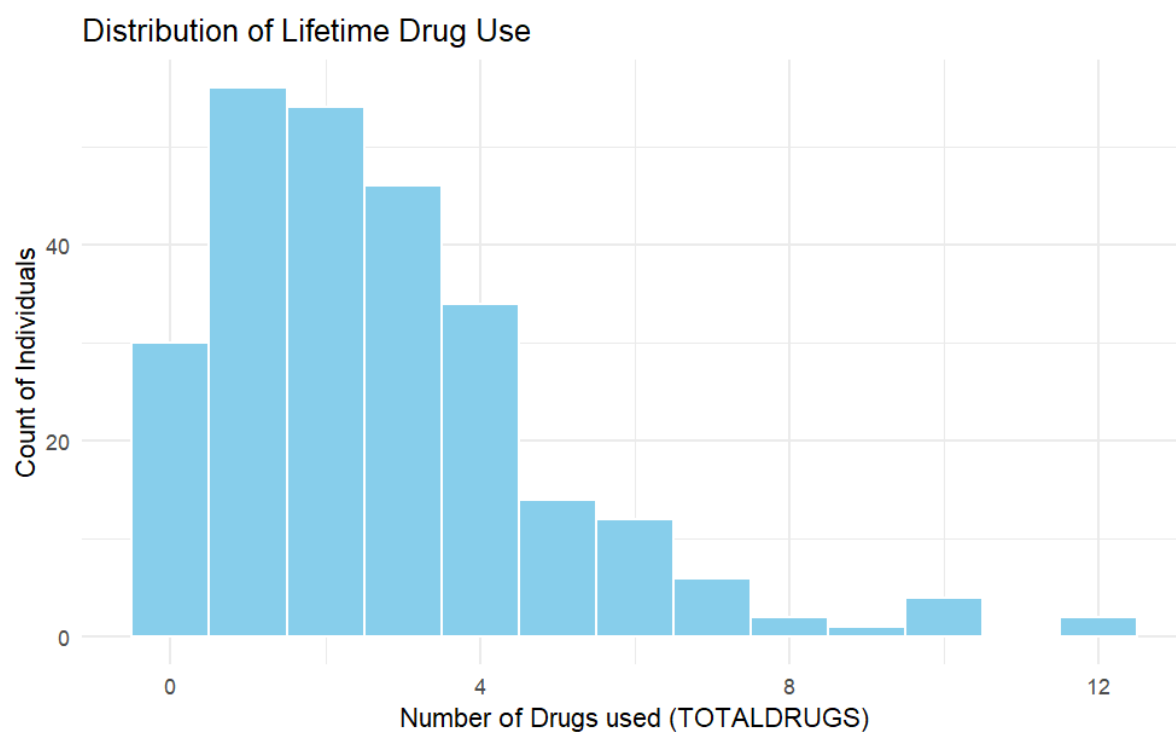


Figure 1 : Bar chart of the distribution of Total drugs

Demographic Distributions

The data set is composed by 83.14 % of female and 16.85% of male, 21.07% of Asian (API) 14.17% of Black, 19.54% of Mixed Race and 45.21% of White. All of these people, 27.58 % are currently enrolled in School and 72.41% are not are shown in the figure 2.

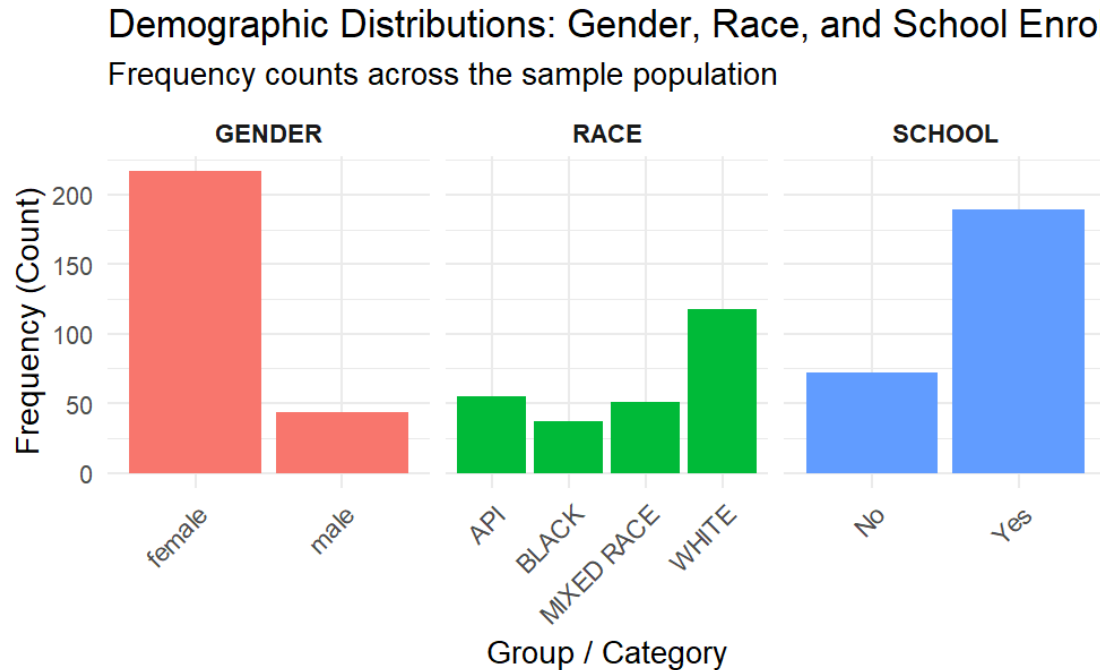


Figure 2 : Repartition of the different groups in the dataset

Lifetime Drug use in Relation to gender

When analyzing the figure 3 we can notice a that the median number of drugs used is higher for males than for females. The male median sits at approximately 4, while the female median is lower, at 2. Males show a slightly tighter middle 50% of data (roughly between 3 and 4.5) compared to females (roughly between 1 and 3). Both distributions appear right-skewed, as indicated by the long upper whiskers and there are several high-value outliers.

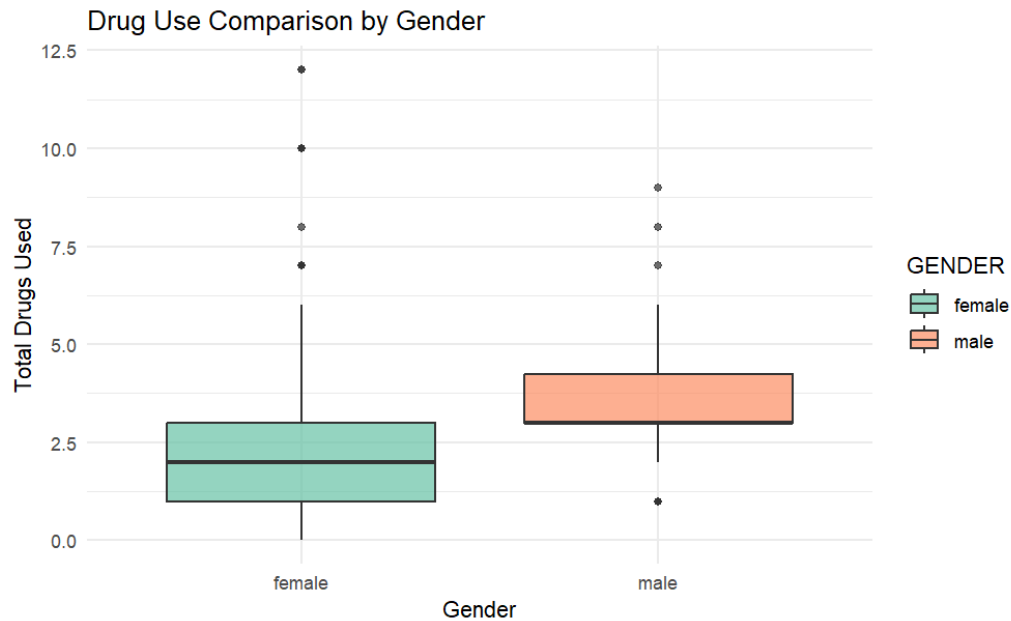


Figure 3 Boxplot of Lifetime Drug use by gender

Lifetime Drug use in Relation to Race

All four racial groups have an identical median of 2. This suggests that, on average, the typical number of drugs used does not differ significantly by race. The boxes (representing the middle 50% of the data) are virtually the same across all groups, spanning from 1 to 4. This indicates that the core distribution of drug use is remarkably consistent across these categories.

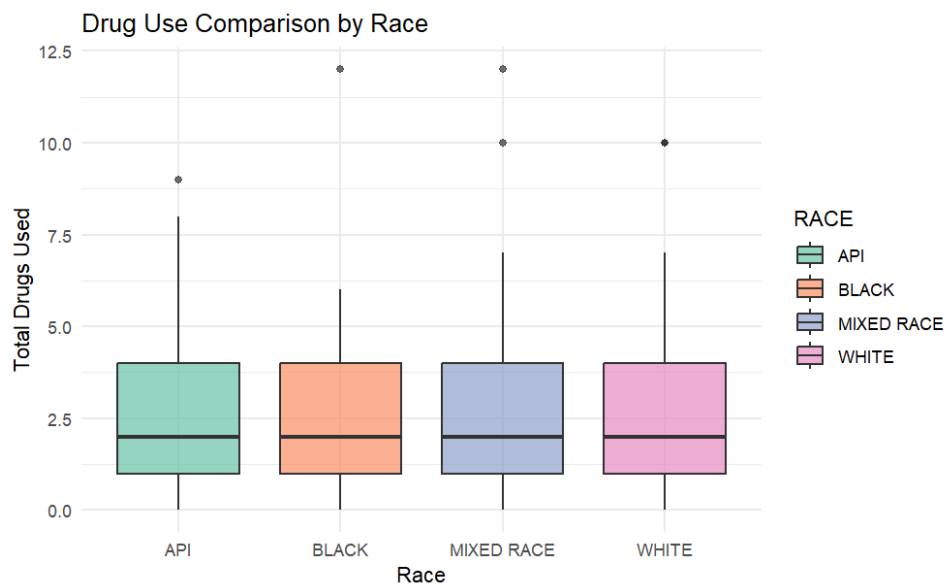
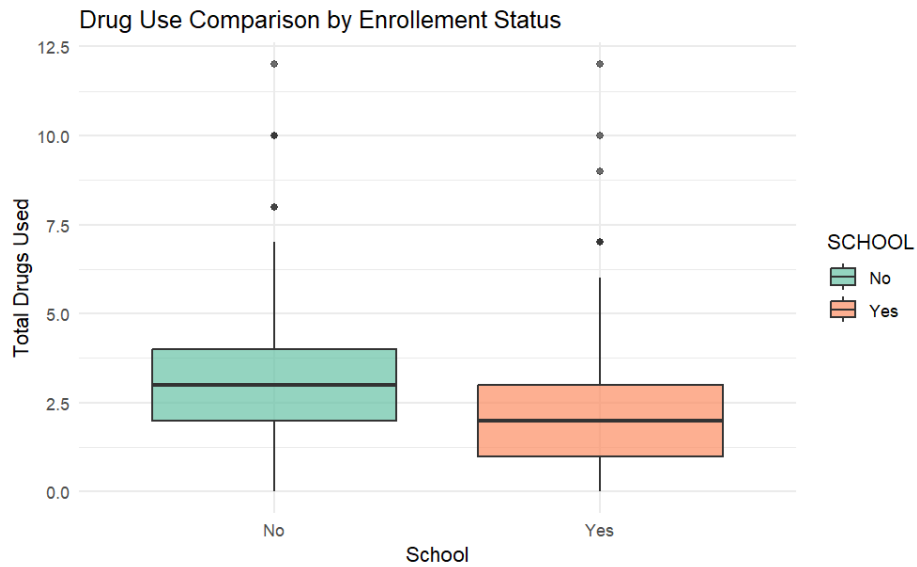


Figure 4 Boxplot of the Lifetime Drug use by Race

Lifetime Drug use in Relation to School enrollement

Those not enrolled in school ("No") have a higher median drug use , compared to those enrolled in school ("Yes"). This indicates that individuals not in school tend to report a higher number of drugs used on average.



CONCLUSION

The study indicates that gender and enrollment status are the primary differentiators of drug use frequency within this dataset, while race appears to have little to no correlation with the typical number of drugs used. Males and individuals not currently enrolled in school represent the subgroups with the highest typical usage rates. Conversely, being female and being enrolled in school are associated with lower median counts. The consistent right-skewness and presence of outliers across all categories suggest that the data does not follow a normal distribution. For future modeling, a Poisson or Negative Binomial regression would be most appropriate to account for the count-based nature of the "Total Drugs Used" variable.

R CODE

Bar chart of the distribution of Total drugs	<pre>ggplot(df, aes(x = TOTALDRUGS)) + geom_bar(binwidth = 1, fill = "skyblue", color = "white") + theme_minimal() + labs(title = "Distribution of Lifetime Drug Use", x = "Number of Drugs used (TOTALDRUGS)", y = "Count of Individuals")</pre>
TOTALDRUGS by GENDER (Boxplot)	<pre>ggplot(df, aes(x = GENDER, y = TOTALDRUGS, fill = GENDER)) + geom_boxplot(alpha = 0.7) + theme_minimal() + scale_fill_brewer(palette = "Set2") + labs(title = "Drug Use Comparison by Gender", x = "Gender", y = "Total Drugs Used")</pre>
TOTALDRUGS by Race (Boxplot)	<pre>ggplot(df, aes(x = RACE, y = TOTALDRUGS, fill = RACE)) + geom_boxplot(alpha = 0.7) + theme_minimal() + scale_fill_brewer(palette = "Set2") + labs(title = "Drug Use Comparison by Race", x = "Race", y = "Total Drugs Used")</pre>
TOTALDRUGS by Enrollement Status(Boxplot)	<pre>ggplot(df, aes(x = SCHOOL, y = TOTALDRUGS, fill = SCHOOL)) + geom_boxplot(alpha = 0.7) + theme_minimal() + scale_fill_brewer(palette = "Set2") + labs(title = "Drug Use Comparison by Enrollement Status", x = "School",</pre>

	y = "Total Drugs Used")
Create the combined plot	<pre> plot_data <- df %>% dplyr::select(GENDER, RACE, SCHOOL) %>% pivot_longer(cols = everything(), names_to = "Category", values_to = "Response") ggplot(plot_data, aes(x = Response, fill = Category)) + geom_bar(show.legend = FALSE) + # Create bar charts facet_wrap(~Category, scales = "free_x") + # Split into 3 panels; "free_x" adjusts labels for each theme_minimal(base_size = 14) + # Modern, clean theme labs(title = "Demographic Distributions: Gender, Race, and School Enrollment", subtitle = "Frequency counts across the sample population", x = "Group / Category", y = "Frequency (Count)") + theme(axis.text.x = element_text(angle = 45, hjust = 1), # Tilt labels so they don't overlap strip.text = element_text(face = "bold")) </pre>
Descriptive Statistics of Total Drugs	<pre> summary_stats <- df %>% summarise(Mean_Drugs = mean(TOTALDRUGS, na.rm = TRUE), Median_Drugs = median(TOTALDRUGS, na.rm = TRUE), SD_Drugs = sd(TOTALDRUGS, na.rm = TRUE), Min_Drugs = min(TOTALDRUGS, na.rm = TRUE), Max_Drugs = max(TOTALDRUGS, na.rm = TRUE)) print(summary_stats) </pre>